

Anthony Rojas Fuentes

Data Automation & AI Engineer

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Summary

Data Automation & AI Engineer specialized in optimizing data pipelines, pioneering AI algorithms, and building robust production-grade software. Hands-on experience on Data Engineering, AI Research and Development, Software Architecture Design and Full-Stack Development. Recognized for rigorous algorithmic knowledge and problem-solving as a **2024 ICPC World Finalist**. Proven track record of translating complex theoretical research into efficient, scalable solutions, backed by international experience developing and testing a custom AI framework at **CERN** and **publishing a paper** presented as **poster at the BIP 2025 IEEE conference**. Passionate about tackling challenging algorithmic problems, automating workflows, optimizing systems, and turning theoretical research into efficient, innovative solutions.

Skills

- **Languages:** Python, SQL, Java, Kotlin, Bash, C, C++, C#, Rust, JavaScript, TypeScript, CSS, HTML, ASM, Markdown, LaTeX
 - **Data Engineering:** Apache Airflow, DBT (Data Build Tool), Apache Spark, NumPy, Pandas, ELT, ETL, Pipeline Design, Data Cleaning, Data Modeling, Automation
 - **Cloud Services:** GCP (Google Cloud Platform), BigQuery, AWS (Amazon Web Services), Azure
 - **AI & ML:** AI, ML, Agents, Agents Skills, MCP, Prompt Engineering, HuggingFace, PyTorch, TensorFlow, Keras, Scikit-Learn, Generative Models, GANs, VAEs, LLM API Integration, Custom Loss Functions, Scientific Computing
 - **BI & Analytics:** Power BI, Tableau, Excel, Jupyter, Matplotlib, Seaborn, Plotly, Data Visualization.
 - **Databases:** PostgreSQL, MySQL, SQL Server, MongoDB, Oracle, Redis
 - **Full-Stack:** Flask, FastAPI, Spring Boot, JWT, Selenium, Selene, React, Vite, Node.js, NPM, Backend, Frontend, Web Development, Zero-config Development, REST APIs
 - **DevOps & Infrastructure:** Docker, Git, CI/CD, GitLab CI/CD, GitHub Actions, Linux, Shell Scripting, Network Configuration
 - **Methodologies:** Agile, Scrum, Kanban, Data Driven Development, Specification Driven Development, AI Assisted Development
 - **Soft Skills:** Effective Communication, Leadership, Complex Problem Solving, Teamwork, Quick Learning, Continuous Learning, Critical Thinking, Creative Thinking, Innovation, Resilience, Adaptability
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Experience

Data Engineer · CriticalRiver Inc., Costa Rica

Mar 2026 – May 2026

- Engineered **Zero-config** data pipelines utilizing **Redis** for real-time streaming to **Azure** storage, alongside implementing **DBT** transformation models for a reproducible analytics layer.
- Designed secure **REST APIs** via **Flask** and **FastAPI** connected to **SQL Server**, and implemented **SAS** and **JWT** secured artifact management flows within **Agile** cross-functional teams.
- Developed automated web testing workflows using **Selenium** and **Selene**, integrating **Redis** to stream execution data and route system logs to **Azure** for long-term auditability.

AI Researcher & Developer · CERN – LHCb Experiment (CONARE Academic Internship), Remote

Jan 2025 – Dec 2025

- Developed **GANU-Lab**, a custom PyTorch framework to orchestrate the training and testing of **state-of-the-art GAN-based generative models** that accelerate LHCb particle physics simulations.
- Enhanced model reliability and interpretability in high-variance environments by researching and implementing **custom loss functions** and **uncertainty quantification** metrics.
- Co-authored a scientific paper and presented the research as a poster at the **7th BIP 2025 IEEE conference**.

Software Engineer & Project Lead · Asociación Pro Adulto Mayor de Naranjo (APAMAN), Costa Rica

Jan 2025 – Jun 2025

- Architected a full-stack healthcare management platform (**Java, Spring Boot, React, PostgreSQL, Docker, AWS**) to digitize legacy workflows into a centralized, modular web ecosystem.
- Engineered EMR modules implementing **RBAC security** to protect patient records, alongside a custom financial automation engine for quota tracking and billing.

IT Infrastructure & Support Engineer · Tecnológico de Costa Rica (SETIAL), Costa Rica

Jan 2023 – Dec 2025

- Developed and deployed internal automation projects to streamline department operations, using CI/CD pipelines and managing the maintenance of computer labs and servers for the Computer Engineering Unit.

Education

B.S. in Computer Engineering · Tecnológico de Costa Rica (TEC) · Jan 2022 – Dec 2025

Certifications & Continuing Education

- **Data Engineering Bootcamp** · CriticalRiver Inc.
- **Transformational Leadership and Soft Skills Program** · Maxwell Leadership Foundation
- **Scrum Fundamentals** · Scrum Academy

Awards & Recognition

- **ICPC World Finalist 2024 & Regional Champion** — Represented Central America at the ICPC World Championship in Astana, Kazakhstan, competing against top global universities in advanced algorithmic problem-solving as Regional Champion (Team “TEC-Silla”).
- **BIP 2025 IEEE Presenter** — Selected for poster presentation at the 7th IEEE International Conference on BioInspired Processing for research on AI-driven simulation and output uncertainty quantification methods at CERN.
- **Vice President, ASODEC-AL** (Computer Engineering Student Association - Alajuela, TEC) — Elected for two consecutive terms; managed academic events and student-faculty liaison initiatives (2024–2025).

Languages

- **Spanish** — Native
- **English** — Full Professional Proficiency